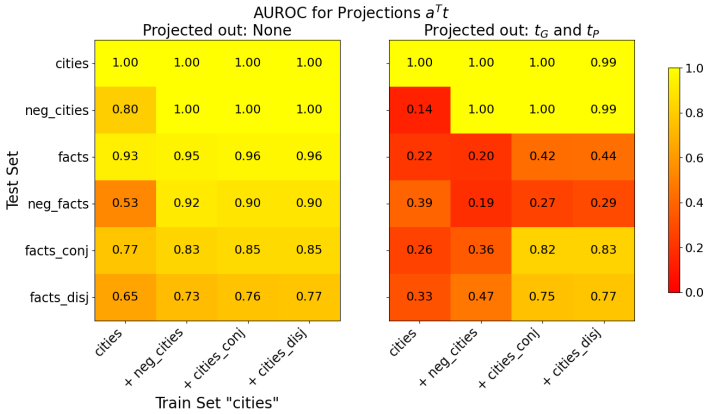

Figure 5: Generalisation accuracies of truth directions $\mathbf{t}$ before (left) and after (right) projecting out $\text{Span}(\mathbf{t}_G, \mathbf{t}_P)$ from the training activations. The x-axis is the training set and the y-axis the test set.

or on the full test set otherwise. The displayed AUROC values are averaged over 10 training runs with different train/test splits. We make the following observations: Left panel: (i) A truth direction $\mathbf{t}$ trained on affirmative statements about cities generalises to affirmative statements about diverse scientific facts but not to negated statements. (ii) Adding negated statements to the training set enables $\mathbf{t}$ to not only generalize to negated statements but also to achieve a better separation of logical conjunctions/disjunctions. (iii) Further adding logical conjunctions/disjunctions to the training data provides only marginal improvement in separation on those statements. Right panel: (iv) Activations from the training set `cities` remain linearly separable even after projecting out $\text{Span}(\mathbf{t}_G, \mathbf{t}_P)$. This suggests the existence of topic-specific features $\mathbf{f}_i \in \mathbb{R}^d$ correlated with truth within individual topics. This observation justifies balancing the training dataset to include an equal number of statements from each topic, as this helps disentangle $\mathbf{t}_G$ from the dataset-specific vectors $\mathbf{f}_i$. (v) After projecting out $\text{Span}(\mathbf{t}_G, \mathbf{t}_P)$, a truth direction $\mathbf{t}$ learned from affirmative and negated statements about cities *fails* to generalize to other topics. However, adding logical conjunctions to the training set restores generalization to conjunctions/disjunctions on other topics.

The last point indicates that considering logical conjunctions/disjunctions may introduce additional linear structure to the activation vectors. However, a truth direction $\mathbf{t}$ trained on both affirmative and negated statements already generalizes effectively to logical conjunctions and disjunctions, with any additional linear structure contributing only marginally to classification accuracy. Furthermore, the PCA plot shows that this additional linear structure accounts for only a minor fraction of the LLM's internal linear truth representation, as no significant third Principal Component appears.

In summary, our findings suggest that $\mathbf{t}_G$ and $\mathbf{t}_P$ represent most of the LLM's internal linear truth representation. The inclusion of logical conjunctions, disjunctions and German statements did not reveal significant additional linear structure. However, the possibility of additional linear or non-linear structures emerging with other statement types, beyond those considered, cannot be ruled out and remains an interesting topic for future research.

5 Generalisation to unseen topics, statement types and real-world lies

In this section, we evaluate the ability of multiple linear classifiers to generalize to unseen topics, unseen types of statements and real-world lies. Moreover, we introduce TTPD (Training of Truth and Polarity Direction), a new method for LLM lie detection. The training set consists of the activation vectors $\mathbf{a}_{ij}$ of an equal number of affirmative and negated statements, each associated with a binary truth label τ_{ij} and a polarity p_i, enabling the disentanglement of $\mathbf{t}_G$ from $\mathbf{t}_P$. TTPD's training process consists of four steps: From the training data, it learns (i) the general truth direction $\mathbf{t}_G$, as outlined in Section 3, and (ii) a polarity direction $\mathbf{p}$ that points from negated to affirmative statements in activation space, via Logistic Regression. (iii) The training activations are projected onto $\mathbf{t}_G$ and $\mathbf{p}$. (iv) A Logistic Regression classifier is trained on the two-dimensional projected activations.